



Town Planning

Course Name:	Indian Knowledge System (FE)
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Definition: Town Planning

Town planning is considered as an art of shaping and guiding the physical growth of the town creating buildings and environments to meet the various needs such as social, cultural, economic and recreational etc. and to provide healthy conditions for both rich and poor to live, to work, and to play or relax, thus bringing about the social and economic well-being for the majority of mankind

- ASI has documented town planning structure dating back to 2600 BCE. Sindhu-Saraswathi Civilization or the Indus Valley Civilization – **Harappa, Mohenjodaro and Lothal sites**
- They followed a system of town planning with similarities in their layouts on a rectilinear basis of main east-west routes directed to the citadels and north-south cross routes (Morris 1979).

Ancient India

- It was situated between Indus River and the Ghaggar - Hakra River (Pakistan and North Western India).
- These cities were also the earliest instances of **Grid iron** town planning.
- The city proper consisted of two components— a citadel, built on high ground and a lower city where the majority of the population lived



Harappa & Mohenjudaro

Main features of town planning in Indus valley civilisation:-

- Streets in perfect grid patterns running in North South direction with intersection at right angles
- World's first sanitation system. Separate covered drains along the streets for waste water.
- All the houses had access to water and drainage facilities. There was Underground sewage and drainage facilities from houses
- Impressive dockyards ,granaries, warehouses, brick platforms and protective walls.
- Massive citadels protected the city from floods and attackers.

Harappa & Mohenjodaro TP

- Houses opened to inner courtyards and smaller lanes.
- Buildings were of masonry construction by Sun dried bricks.
- Small houses ranging from 2 rooms to mansions with many rooms were built
- Streets within built up areas were narrow.
- Distinct zoning for different groups.
- Public Structure like Great Bath was built. Helical pumps were used for pumping water in Great bath.
- Principal buildings like monastery & bath – indicated religious culture.

Harappa & Mohenjodaro TP

Vedic period: (400 BCE)

- In this period, Vedas as well as books were wrote on town planning.
- In “Vishwa-karmaprakash” it was stated that “first layout the towns and then plan the houses.”
- “Shilpshastra” wrote by “Architect mansara” discussed study on soil, topography, climatology and various layouts like dandaka, swastika, padmaka, nandyavarta. dandika style nandyavarta style
- **Buddhist period: (up to 320 CE)** :During the period of emperor chandraguptamaurya, kautilya (chanakya) was the chief minister who wrote the famous “arthashastra”, a treatise of town planning

Ancient Texts on TP

- Intensive training given to ‘Sthapati’ (architect or town planner)
- Towns usually oblong shape, generally near river banks, sea shore or large lakes.
- Typical town consisted of market, streets, public buildings, residences for people, palaces, temples, recreation centres, gardens, tanks, underground passages, city forts etc.
Eg: Ayodhya, Patliputra (*present patna*), Takshasila, Nalanda (*early University towns*)

Ancient Towns



Ayodhya and Patliputra

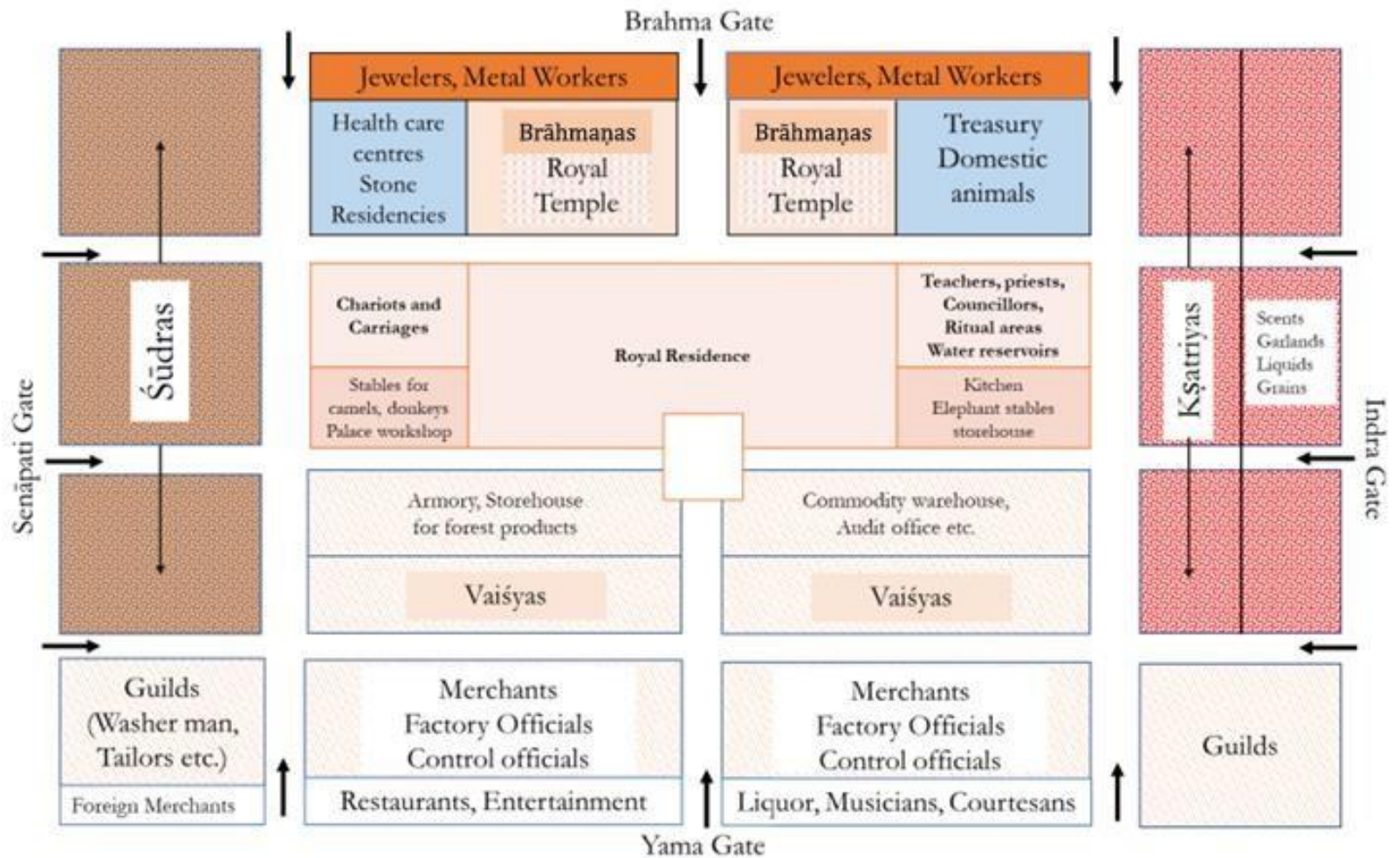
Planning of a town is dependent on the various factors:

- Soil Type
- Climatic Conditions
- Topography
- Wind Orientation
- Orientation to Take Advantage of Sun and Wind

Factors of Town Planning

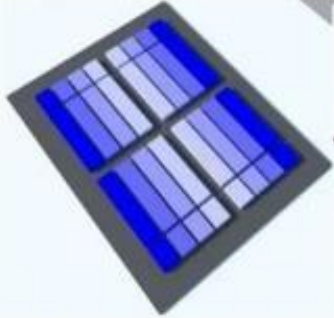
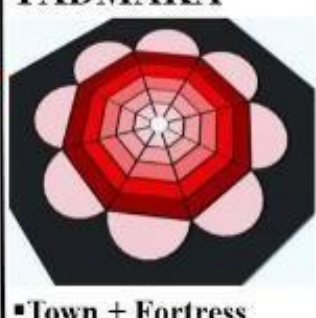
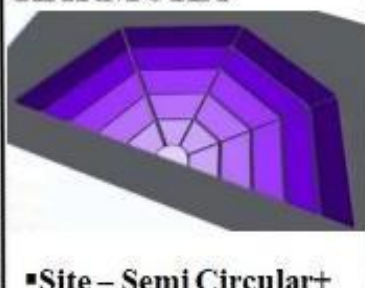
- The towns were highly influenced by the Site conditions.
- The towns were generally located along the bank of the water body.
- A flowing stream was preferred for Sanitary requirements.
- The towns on the river edge were OBLONG shape; to take maximum advantage of the river.
- Main Street (King/ Raja Marg) were aligned East-West to get roads purified by the Sun's rays; while the shorter roads were along North — South.

Features of Town Planning



* The crematoriums are outside the city at designated places (not shown in the figure)

FIGURE 12.1 A Schematic Representation of Kautilya's Plan for the Fortified City

DANDAKA  ▪ Small Village – Square + Rectangular. ▪ Gates On all sides.	SARVATHOBADRA  ▪ Large Village & Town. ▪ Square Site. ▪ Temple is in centre.	NANDYAVARTA  ▪ No Village Only TOWN. ▪ Circular/ Square Site. ▪ Temple is in centre. ▪ Nandyavarta - Flower	PADMAKA  ▪ Town + Fortress around ▪ Plan – Lotus Petals ▪ Island + MOAT
SWASTIKA  ▪ Site – Any shape ▪ Diagonal Streets ▪ Rampart Wall + MOAT ▪ Inspiration Swastika ▪ Temple - Centre	PRASTARA  ▪ Site – Either Square or Rectangle ▪ Segregation for Rich And Poor	KARMUKA  ▪ Site – Semi Circular+ Parabolic + At River Bank+ Bow Shaped ▪ Streets At Right Angle Forming Various Blocks	CHATURMUKHA  ▪ Largest Town to Smallest Village ▪ Site – Square + Rectangle ▪ Temple - Centre

Ancient town classification

TABLE 12.4 Alternative Designs for Town Planning Found in Mānasāra

<i>Nature of Designs</i>			
<i>Design</i>	<i>Shape</i>	<i>Street and Roads</i>	<i>Other Details</i>
Daṇḍaka	Stick/staff/ phalanx	Parallel set of straight streets (one to five) crossing each other at right angles.	Good for priests, Sages, intellects, etc. No. of houses: 12–300.
Nandyāvarta	Square or oblong	It should have one to five carriage roads together with the surrounding street. Internal roads have one footpath and the outer two footpaths.	
Sarvatobhadra	Square or oblong	Number of car streets varies from 1 to 5.	Town/village is secured by a wall and ditch with four large gates on the sides and as many on the corners.
Padmaka	Lotus shaped	Length and breadth are equal while the enclosing walls can be circular, quadrangular, hexagonal, or octagonal. There are 4 to 8 streets lined with the houses.	According to Mayamata, there are five varieties of this design. Gates are placed in four cardinal directions.

Svastika	Svastika	Two streets passing through the center. Traversing streets are planted in the clockwise direction. The outermost roads should be lined with a single row of houses and other streets with a double row of houses.	Temple at the center, has eight gates (two each on every side).
Prastara	Square or oblong	Space needs to be divided into 4, 9, 16 wards by a network of streets. Roads to be constructed like a chessboard system.	The village needs to have enclosed walls and ditches with four principal gates on the south and subsidiary ones in the corners.
Kārmuka	Semi-circular like a bow	2 car streets, 1 principal street. The number of traverse streets can be from 1 to 5. Houses range on both sides of the streets.	This type of design is best for a seashore. The number of gates is optional.
Caturmukha	Square or oblong	4 car streets on four sides. Two large streets crossing at right angles in the center dividing the whole site into four blocks. Each ward must have four smaller roads.	It is specially meant for traders.

- Here are the principles of ancient Indian town planning that are still relevant today:
 1. Grid-based layout – Ensures easy navigation, systematic growth, and better traffic management.
 2. Efficient drainage systems – Prevents waterlogging and maintains urban hygiene.
 3. Zoning of spaces – Separation of residential, commercial, and public areas avoids congestion.
 4. Use of local materials – Encourages eco-friendly and sustainable construction.
 5. Orientation and ventilation – Houses planned for natural light and air reduce energy use.
 6. Public spaces & utilities – Provision of wells, tanks, and community areas enhances social life.
 7. Security & resilience – Fortified designs remind us of the need for safe and disaster-resilient cities.

Crisp Set of learning from Ancient Indian Town Planning
